

Lab Report 9: Controlling Spider Servos in C++

Earl Chan, Nico Constanti

Chan.ea@northeastern.edu

Constanti.n@northeastern.edu

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Abstract:

In this lab, we used C++ to control a servo motor. We created a motor class. In this class, we created methods to control the delay and angle of the motor. We also implemented methods that allowed us to check if a motor was ready for another move command. After creating all these methods, we implemented a SpiderLeg class which held three motors, one for the hip, one for the knee, and one for the ankle. This allowed us to control spider legs in an organized manner.

Introduction:

The main point of this lab was to reinforce our knowledge of object-oriented programming, using object methods on references to the object, and using C++ to manipulate hardware by using the output pins as registers.

Pre Lab:

For this prelab, we first needed to create a table showing the conversion between an angle, duty cycle in ms, and cycle count. This table included the angles of -90, 0, and 90 degrees.

Angle (Degrees)	Duty Cycle (ms)	Duty Cycle (50MHz Ticks)
-90	.05	25000
0	1.5	75000
90	2.5	125000

We also needed an algorithm to make this conversion from angle to clock cycles.

$$\text{clock cycles} = 25000 + 1250(\text{angle} + 90)$$

Lastly, for this prelab, we needed an equation to translate a speed between 0 to 100, to a delay between 1000 and 1700 cycles, in this equation, we want the lowest speed to have the highest delay, and a speed of 100 to have the lowest delay.

$$(700 - (\text{speed} * 7) + 1000) = \text{cycles}$$

Results:

Assignment 1:

For assignment 1, we needed to make three separate servos on the spider robot to turn in an organized manner on a button press.. To do this, we first implemented Move() and SetSpeed() methods in the Motor class. We implemented these methods by first implementing limit checking, and then by using the WriteRegister() methods we made in the first labs. After this, we created Reset(), IsReady(), GetfAngle(), and GetfSpeed() methods by reading the information stored in the class properties by our previous functions. Lastly, we implemented a main file that read the buttons using our ReadButton() method that we implemented in a previous lab. When these buttons are pressed, we used our motor object, to move three motors at 15 degree increments.

Assignment 2:

For assignment 2, we needed to make implement a SpiderLeg class which would allow us to move a spider's leg. To do this, we implemented the move Joint method which allowed us to move one of the three motors in the spider leg. Using this we were able to cycle through several motions that a Spider would make

Conclusion:

In this lab report, we solidified our understanding of object oriented programming